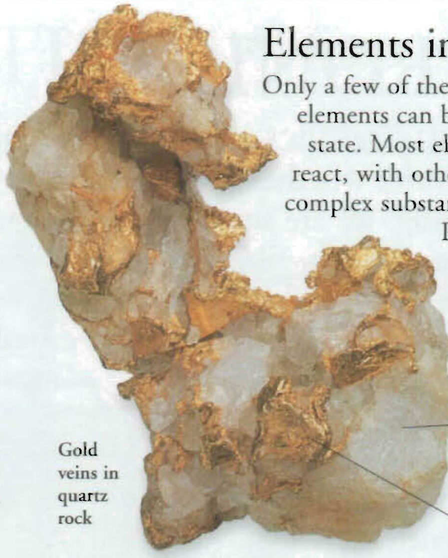
# ELEMENTS


AN ELEMENT IS a substance composed of only one type of atom. Elements are the most basic substances in the Universe and cannot be split into anything simpler. There are 109 elements – 91 of which occur naturally, and 18 of which can be made artificially. All life on Earth is based on the element carbon, which is vital to the functioning of living cells. Oxygen is the most plentiful element on Earth. It occurs in air, water, and even rocks.

## Elements in nature

Only a few of the naturally occurring elements can be found in their pure state. Most elements combine, or react, with other elements to form more complex substances called compounds. Pure gold can be mined directly from the ground because it is unreactive – that is, it does not readily form compounds.



*Quartz rock is a compound of the elements silicon and oxygen.*

*Pure gold.*

**Gold veins in quartz rock**

## Groups of elements

Just as the members of a human family share the same characteristics, there are "families" of elements that have similar properties. An element's chemical properties are determined by the structure of its atoms. Elements in the same group have similar atomic structures.

*Calcium gives bones their hardness.*

### Alkaline-earth metals

Calcium and magnesium belong to the group of elements called the alkaline-earth metals. They are so named because they form alkaline solutions in water, and their compounds occur widely in nature. Calcium, for example, occurs in sea shells, bones, teeth, milk, and chalk. Magnesium occurs in the substance chlorophyll, which plants use to make food by photosynthesis.

**Iodine** **Bromine** **Chlorine**

### Halogens

Swimming pools smell the way they do because the halogen chlorine is put in the water to kill germs. Compounds of fluorine, another halogen, are put in water and toothpaste to prevent tooth decay. The halogens, which also include iodine, bromine, and astatine, are all strong-smelling, highly reactive non-metals.

### Alkali metals

Potassium (which is used in fertilizers) and sodium (which occurs in salt) are both alkali metals. All the elements in this group are soft, extremely reactive metals. They react violently or even explosively with water to form alkaline solutions.

**Reaction of potassium in water**

*Iron compounds are often red, black, or brown.*

*Iron sulphide*

*Iron carbide*

*Iron oxide*

**Coloured compounds of iron**

### Transition metals

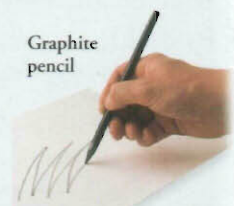
The transition metals are a large group of hard, dense elements that conduct electricity and heat well, form coloured compounds, and some of which (iron, cobalt, and nickel) are magnetic. Other transition metals include copper, gold, chromium, titanium, platinum, and tungsten.

### Noble gases

Multi-coloured street signs often contain noble gases, because each of these gases glows a different colour when electricity flows through it. Neon, for example, glows red, helium yellow, and argon blue. The noble gases are unreactive non-metals that rarely form compounds.

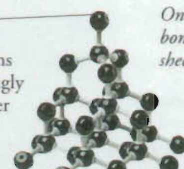
### Allotropes

It may seem difficult to believe, but hard, sparkling diamond is made of the same types of atoms as soft, black graphite. Diamond and graphite are allotropes of carbon, meaning that they are different physical forms of the same element. Their atoms link up in different ways to make them look and behave differently.

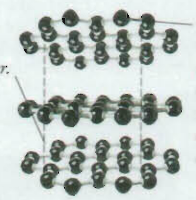


**Graphite pencil**

**Diamond** consists of carbon atoms linked strongly to each other in a rigid framework.



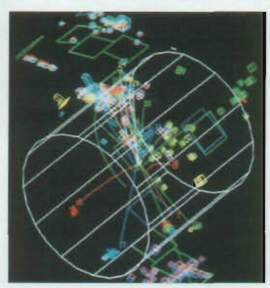
*Only weak bonds hold sheets together.*



**Graphite** is made up of sheets of carbon atoms that can slide over each other easily.

## Artificial elements

New elements can be created by bombarding existing elements with high-speed subatomic particles in a device called a particle accelerator. Since 1937, scientists have made 18 new elements, some of which only exist for a few millionths of a second.



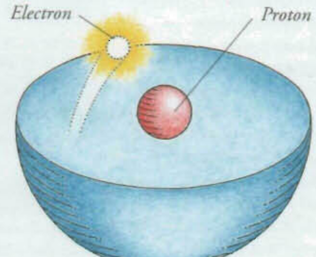
**Computer image of a particle accelerator collision**

### Hydrogen

The element hydrogen makes up 90 per cent of all the matter in the Universe. It was the first element to form when the Universe was created in the explosion known as the Big Bang. Hydrogen is a tasteless, colourless, odourless, non-toxic gas. It is the simplest of all the elements, with atoms containing just one proton orbited by a single electron. Hydrogen gives acids their acidic properties.

*Electron*

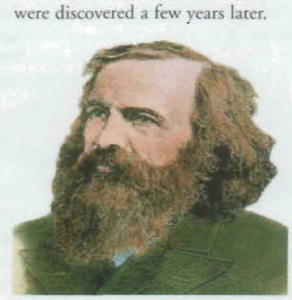
*Proton*



**Hydrogen atom**

## Dmitri Mendeleyev

In 1869, the Russian chemist Dmitri Mendeleyev (1834–1907) devised a chart called the periodic table, which classified the 63 elements then known into different groups. He used the table to predict the existence of three new elements, all of which were discovered a few years later.

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